HACK APERTUS 2026 · TRACK 2A

See It, Say It, Pick It

A local vision-language pipeline for safe first-pick grounding and ordered robot actions

Team Nightcats



02

Model responsibilities

SEE

Qwen2.5-VL

objects + boxes

MASK

SAM 2.1

pixel masks

REASON

Apertus

language + plan

CHECK

Safety rules

one repair

GROUND

Geometry

interior (u, v)

Apertus decides meaning and action order. Deterministic modules enforce the physical contract.

Grounding and safety

FIRST PICK

Mask interior beats box center

Padded distance transform selects maximum clearance. Thin shapes fall back to the valid pixel nearest the mask centroid.

One object in the gripper · 85 mm opening
No trays, people, hands, unstable supports, throws or drops



04

Verification status and failure analysis

PASS

full automated suite

50

ordered public instructions

6

challenge categories

Expected failure classes

Missed small objects
Spatial relation mistakes
Planner over-refusal
Mask-edge depth errors

PUBLIC DEV SCORE

24.0 / 65
official evaluator

Held-out output: 40 / 40 IDs

05

Reproduction in Colab

SUBMISSION WORKFLOW

Run colab/Hack_Apertus_2A_Heldout_Colab.ipynb

01	**Upload**	Upload runtime-source.zip and the input-only dataset
02	**Run**	Execute the staged local inference workflow on an A100
03	**Validate**	Check exact instruction IDs and the official result schema
04	**Export**	Download results_heldout.json for submission